



**A Business Intelligence Report on the
Daily Hit Charts from 2017 to 2021 of Spotify**

Far Eastern University - Manila

In Fulfillment of the Requirements in 3rd Year DSC1103- Data Warehousing and Management

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I. Objectives

This Business Intelligence (BI) report utilizes Spotify's global charts dataset, spanning from January 1, 2017, to December 31, 2021, and covering 70 countries. The dataset, sourced from [Kaggle](#), consists of over 26 million rows and 9 columns (3.8 GB), with key metrics such as streams, artists, regions, rankings, and timestamp. It provides a comprehensive view of Spotify's performance globally and regionally, including the Philippines.

Spotify, as one of the leading music streaming platforms, serves as a vital tool for understanding user behavior and trends in the music industry. By analyzing streaming data, we can identify growth patterns, evaluate artist and song development, and uncover market opportunities. The insights gained from this report can guide Spotify's strategies for enhancing user engagement, supporting artists, and optimizing market-specific campaigns.

The specific objective of this Business Intelligence report is to:

1. To examine five years of Spotify data to uncover significant trends in streaming volume and user engagement, with a specific focus on the Philippines
2. To assess the performance and growth of top artists and songs, highlighting key drivers of their success and sustained popularity.
3. To deliver actionable insights to address challenges, if there will be.
4. To pinpoint untapped potential in emerging markets and assess regional growth patterns to guide resource allocation and marketing strategies.
5. To create tailored approaches to promote Filipino artists, expand their global reach, and strengthen Spotify's position in the local market.

II. Tools and Technologies Used

To do the visualization of the Business Intelligence (BI) report, **Python 3** and **Jupyter Notebook** were used as the primary tools, with the dataset accessed through CSV file. The following libraries were utilized to analyze the data: **numpy** and **pandas** for data manipulation and processing; **matplotlib** and **seaborn** for creating visualizations such as graphs and heatmaps; **IPython.display.Markdown** for displaying markdown content within the notebook; **country_converter** for mapping and transforming country names; and **plotly.express** for generating interactive map charts. Additionally, **requests** and **collections.Counter** were used for scraping and analyzing external data, which supports the recommendations section of the report.

III. Methodology

The methodology for this BI report follows a structured approach, encompassing data collection, cleaning, preprocessing, and visualization to generate key insights from the Spotify dataset. Below are the steps taken:

a. Data Collection

- i. The dataset was obtained from Kaggle, consisting of Spotify's global charts data from January 1, 2017, to December 31, 2021. It includes 70 countries and tracks key metrics of streams, artists, regions, rankings, and timestamp. The dataset was accessed in CSV format and loaded into a Python Jupyter Notebook for further analysis.

b. Data Cleaning

- i. The first step in data cleaning involved handling missing values. We identified missing data in the columns **title** and **artist** and decided to drop rows with these missing entries, as these were minor gaps and did not significantly impact the report.
- ii. Additionally, missing streaming data was observed in the **viral50** chart type. Since this was an outlier, records related to that were excluded from the analysis.
- iii. The dataset was then examined for data integrity; regions with less than 95% data consistency were excluded to ensure that the analysis would be accurate and representative. Relevant columns, specifically, **title**, **rank**, **date**, **artist**, **region**, and **streams**, were retained for the final dataset, and data types were adjusted accordingly to optimize performance.

c. Data Processing

- i. In this step, additional columns were created, including year and month, for easier time-based analysis. The dataset was then transformed and processed to prepare it for analysis. We applied various data transformation techniques, including sorting, aggregating, and calculating growth rates, to make it suitable for the specific analysis goals.

d. Data Visualization and Analysis

The visualization and analysis step focused on stream-based data, divided into several key areas to derive insights:

1. Stream Analysis (Global and Regional)

- a. Global Streams: The total streaming volume over the entire dataset period was analyzed to identify overall trends and growth patterns.
- b. Global Growth Rate (Year-on-Year): A comparison of yearly streaming growth to understand whether Spotify's streaming volume is increasing or declining.
- c. Streams by Region: A breakdown of streams per region allowed us to understand the distribution of Spotify usage across different countries and regions.
- d. Market Shares: The market share of each region was calculated to identify which areas contribute the most to Spotify's streaming volume and track changes over time.

2. Artist Analysis (Globally and Locally)

- a. Ranking Most Successful Artists: Global and local artist performance was analyzed by identifying the top artists based on total streams, with a focus on trends over the years.
- b. Most Successful Songs of Best Artist: For the top artists, the most streamed songs were identified, revealing the factors behind their success.
- c. Annual Development of Best Artist: Yearly analysis was performed to observe how the top artists' streaming volumes evolved over time, highlighting the most successful years for each artist.
- d. Region Performance of Best Artist: Regional performance analysis showed where the top artists gained the most streams, identifying key markets for their music.

3. Song Analysis (Globally and Locally)

- a. Top Songs Globally: The most streamed songs globally were identified to understand the preferences of Spotify users.
- b. Average Rank by Region of Best Song: The performance of the top song was analyzed by calculating the average rank in different regions, offering insight into regional song preferences.
- c. Rank Development of Top Song: The rank development of the most popular songs was tracked over time to see how their rankings fluctuated and stayed within the charts.

This approach facilitated strategic key insights and recommendations for both global and local Spotify initiatives.

IV. Application Output

A. Data Preprocessing

```
[49]: region_data_volume = np.round(df['region'].value_counts() * 100 / full_data, 1)
      region_data_volume

[49]: region
      Brazil      99.9
      Argentina  99.8
      United States 99.8
      Austria     99.7
      Australia   99.2
      ...
      Russia      27.6
      Ukraine      27.6
      Bulgaria     26.0
      South Korea   16.3
      Luxembourg    2.5
      Name: count, Length: 69, dtype: float64

[54]: region_data_volume_ph = np.round(df['region'][(df['region'] == 'Philippines')].value_counts() * 100 / full_data, 1)
      region_data_volume_ph

[54]: region
      Philippines  98.2
      Name: count, dtype: float64

      The Philippines is above 95% data integrity.

[57]: len(region_data_volume[region_data_volume >= 95]) / df['region'].nunique()

[57]: 0.5797101449275363
```

Figure 1. Screenshot of Data Integrity Process

40 out of 70 countries in the dataset hold data integrity (incl. PH) .For further analysis on region level, we will only consider regions with data quality of min 95%

Most Streamed Song in a Single Day (Globally)

```
[101]: df[df['streams'] == df['streams'].max()]

[101]:
```

	title	rank	date	artist	region	streams
22999369	Easy On Me	1	2021-10-15	Adele	Global	19749704.0

Most Streamed Song in a Single Day (Locally)

```
[105]: df_ph = df[df['region'] == 'Philippines'] # Filter for PH region
      df_ph[df_ph['streams'] == df_ph['streams'].max()]

[105]:
```

	title	rank	date	artist	region	streams
19899414	Butter	1	2021-05-21	BTS	Philippines	753286.0

Figure 2. Screenshot of Most Streamed Song in a Single Day (Globally & Locally)

Easy on Me by Adele is the top song that was streamed in a single day with over 19 billion streams. While locally, Butter by BTS is the top song in a single day with a little bit over 700 million streams

B. Streams Global Analysis

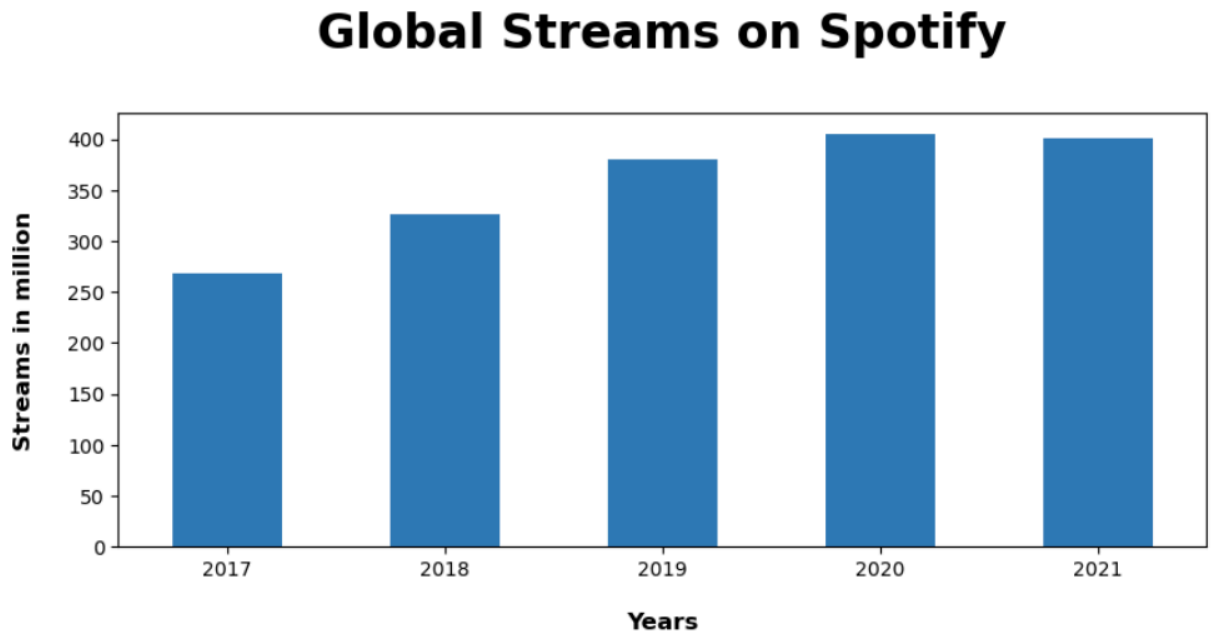


Figure 3. Global Streams on Spotify

Global Growth Rate YoY

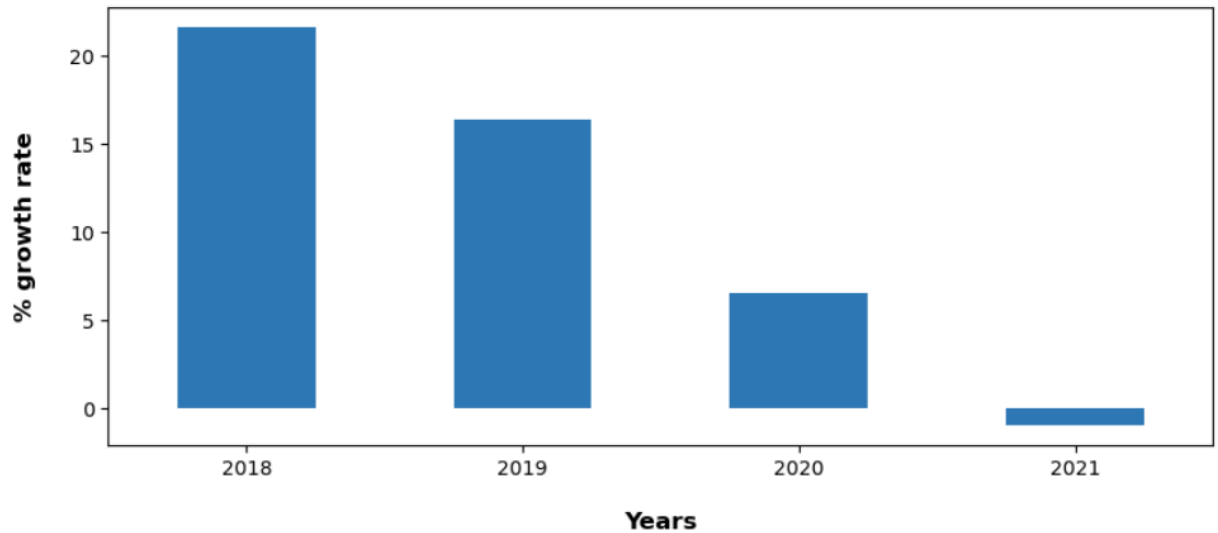


Figure 4. Global Growth Rate Year-On-Year

Spotify streams are continuously increasing until 2020. But the growth rates are decreasing year over year. 2021 is the first year where Spotify is facing an overall decline.

Total Streams per Region by Millions



Figure 5. Map Chart of Total Streams per region by Millions

As can be obviously seen, the map chart did not include all countries as the dataset only has 70 countries. With that, Different countries around North and South America contribute to billions and billions of streams on Spotify, while there is still a significant contribution on the South East Asia countries.

Market Share Development Top 10 Regions

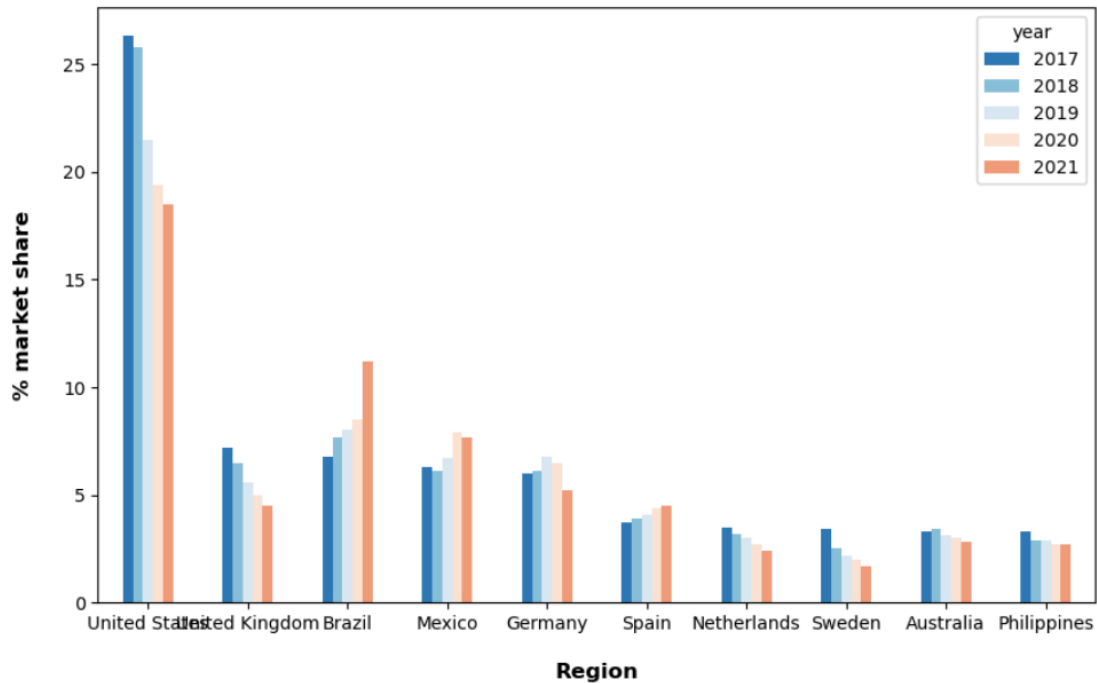


Figure 6. Market Shares Development Top 10 Regions

Most relevant region in 2017 was the USA with an incredible share of 26.3% streaming volume. But between 2017 and 2021, their market share decreases. Specially, Brazil, Mexico, and Spain are expanding the market share.

As we can see also, the Philippines is part of the top 10 regions.

Let's double check this development by calculating year over year growth rates by region.

YoY Growth Rates Top 10 Regions

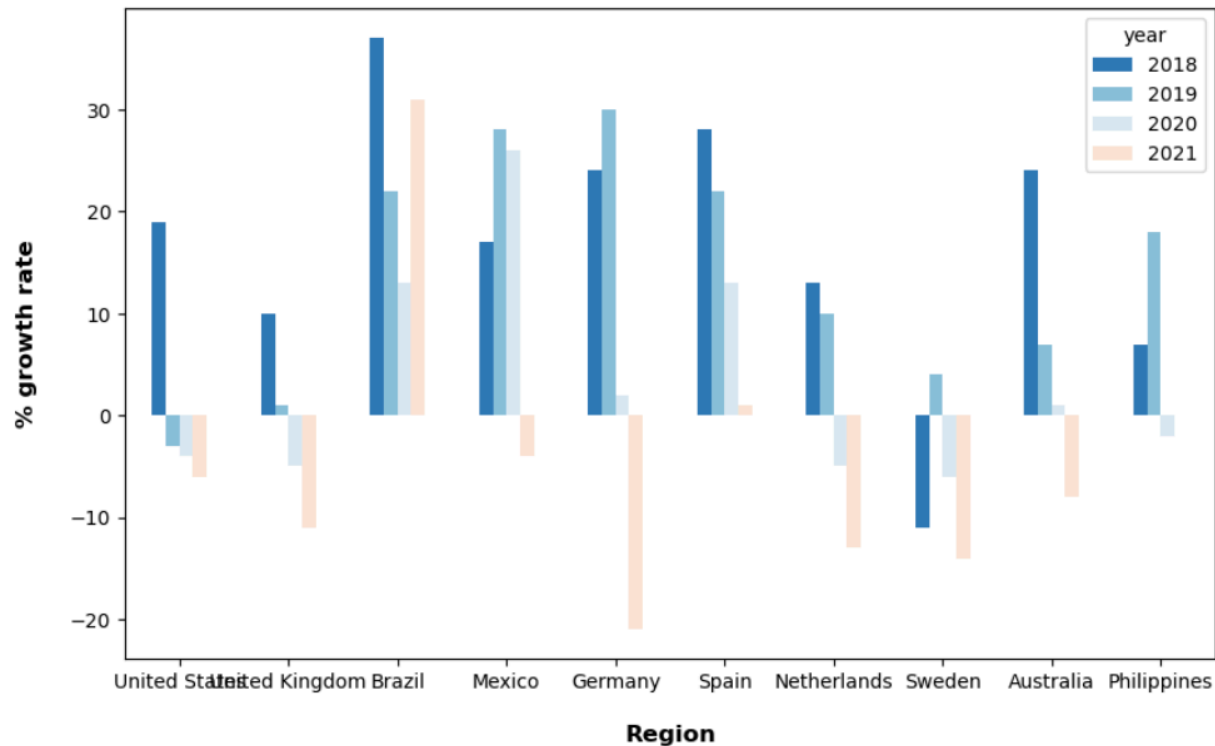


Figure 7. Year-On-year Growth Rates Top 10 Regions

Confirmed by the YoY growth rates, the USA and UK are losing market share because they only show decreasing trend in streaming volume since 2019.

Brazil and Mexico show incredible growth rates >30%. Brazil can even resist the overall downwards trend in 2021 in contrast to all other countries.

Looking at the Philippines, there is a big growth in the year 2018 and 2019, however, there is a decrease in 2020, while being stable in the last year.

% YoY Growth Rates by Region

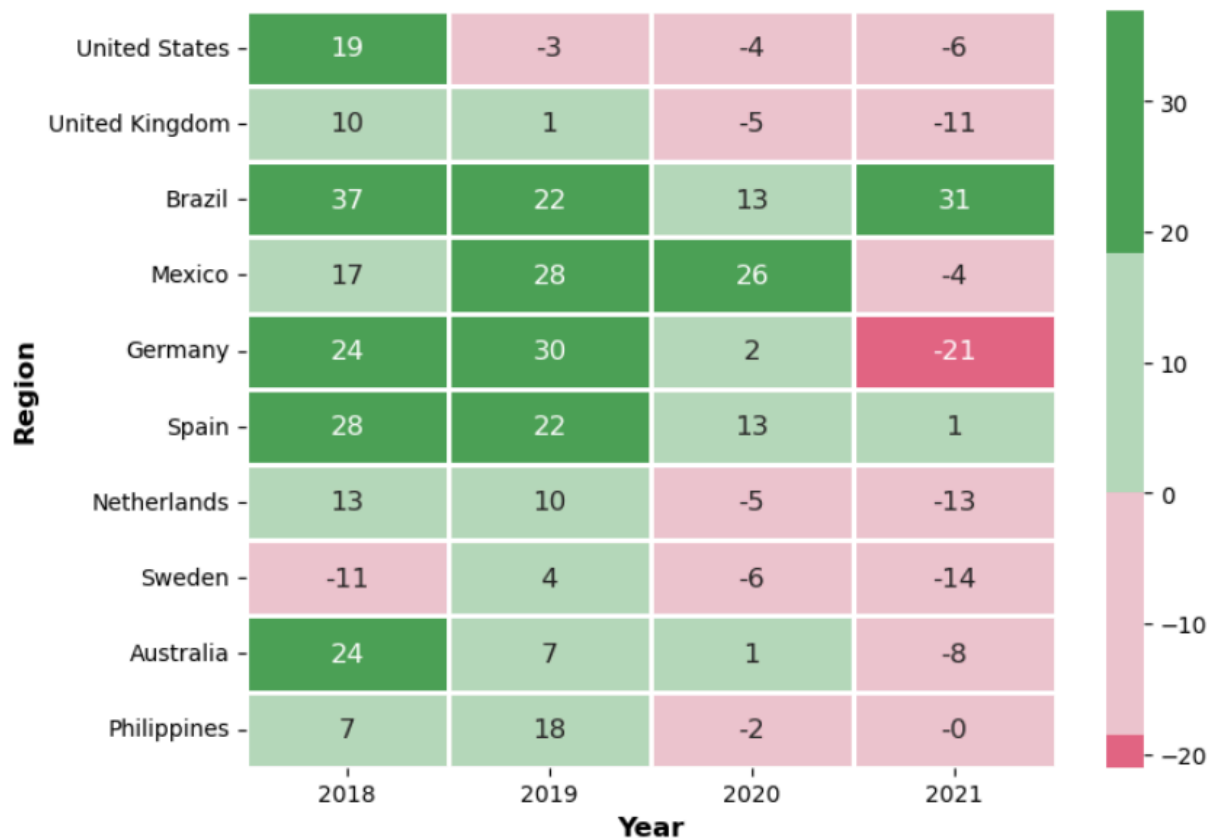


Figure 8. Heat Map of Year-On-Year Growth Rates by Region

Among the top 10 regions, Brazil is the only one which still continues to grow in 2012. All other countries show a decline in the following overall Spotify trend.

Germany shows the worst development in 2021 with 21% reduction. Sweden seems to be the least promising market showing decreases almost every year.

C. Streams Analysis

GLOBALLY

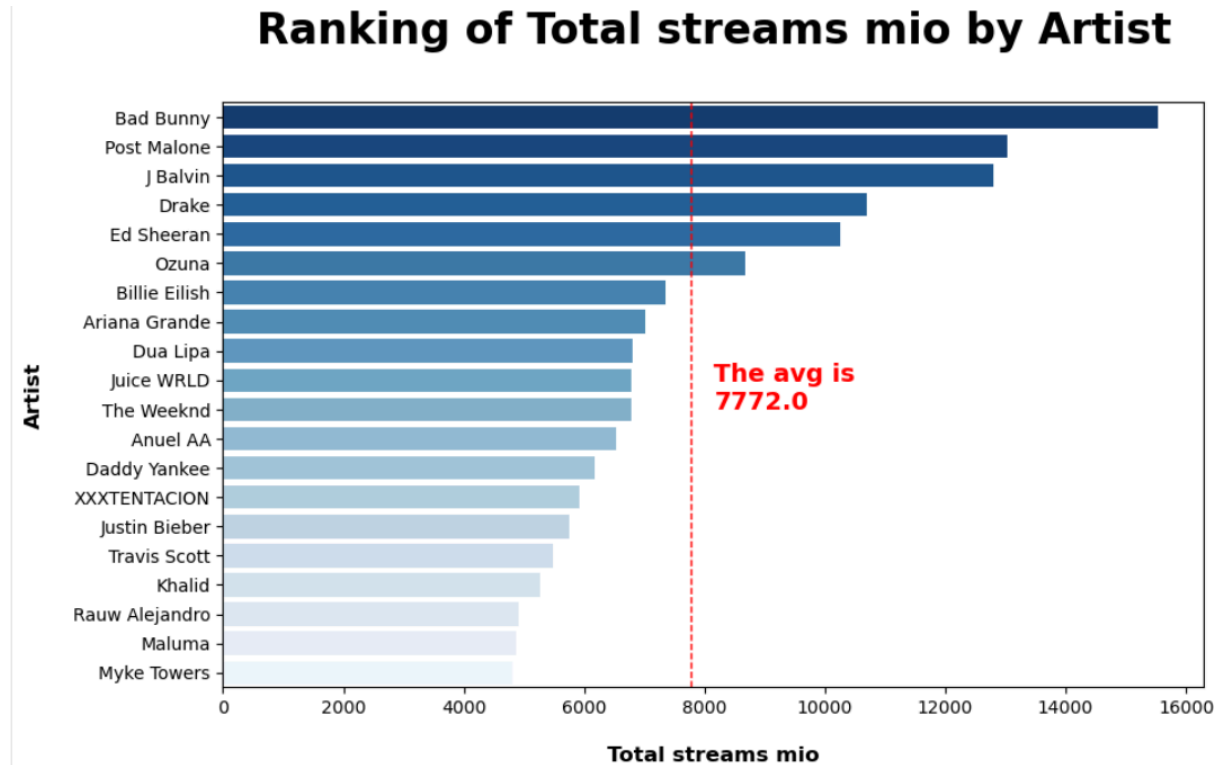


Figure 9. Ranking of Total Streams in Millions by Artist

Measured by total Streamings, "Bad Bunny" is the most successful artist with more than 15.5 billion streams between 2017 and 2021.

Compared to the second artist "Post Malone," there are some significant differences in their key figures. Post Malone has by far the lowest number of songs in the charts. On the other hand, he is leading in the avg streamings per song and the duration in the charts.

Ranking of Total streams mio by Title

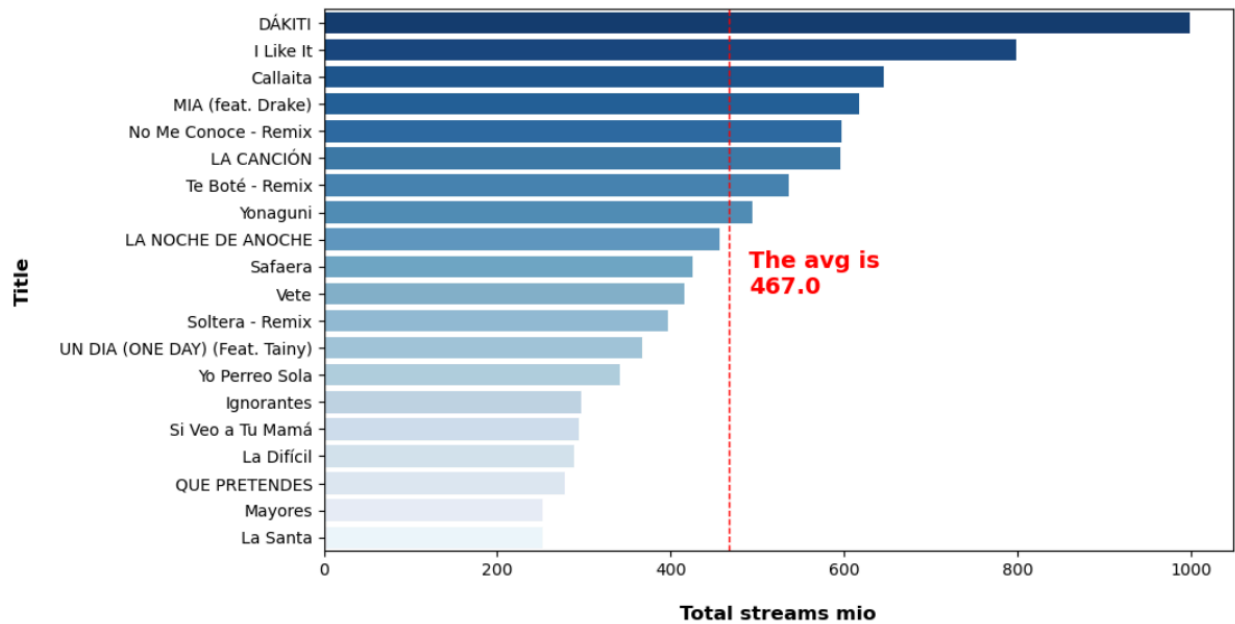


Figure 10. Ranking of Total Streams in Millions by Title of Top Artist

"DAKITI" is the most successful song from "Bad Bunny." It charted in 7087 straight days.

Annual Streams from Bad Bunny

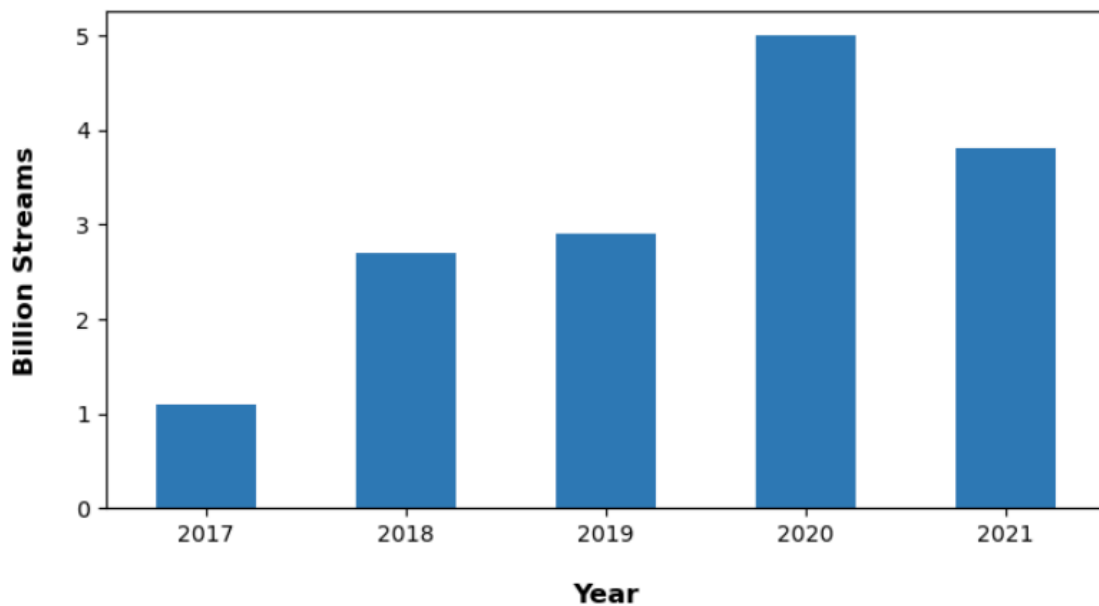


Figure 11. Annual Streams from Bad Bunny

Bad Bunny's most successful year was 2020 with almost 5 billion streams. He had 72 songs in the top 200 and achieved 69 million streams for every song.

LOCALLY

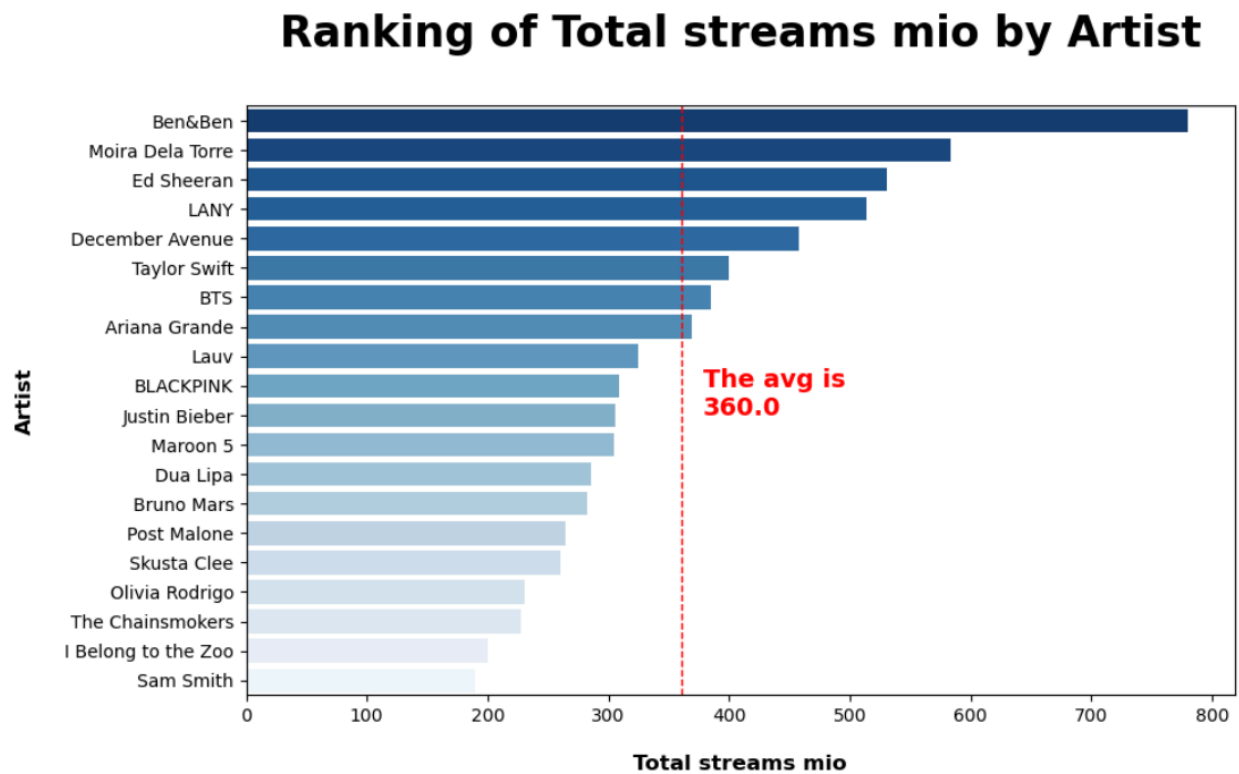


Figure 12. Ranking of Total Streams in Millions by Artists in PH

From the figure above, Ben&Ben ranked the top artist in PH with a great gap with the 2nd artist Moira. The Ben&Ben has almost 800 million total streams.

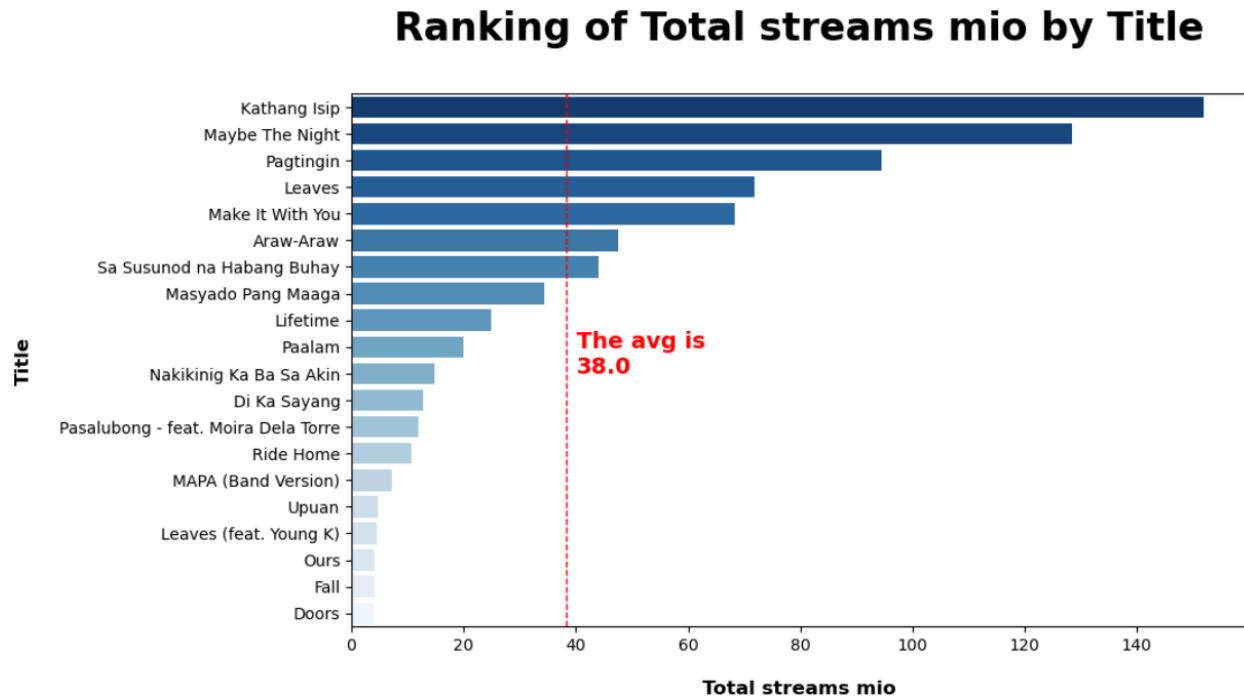


Figure 13. Ranking of Total Streams in Millions by Title of Songs of Top Artist in PH

The top 2 songs from Ben&Ben have reached more than a 100 million total streams, backing up why Ben&Ben was the top local artist.

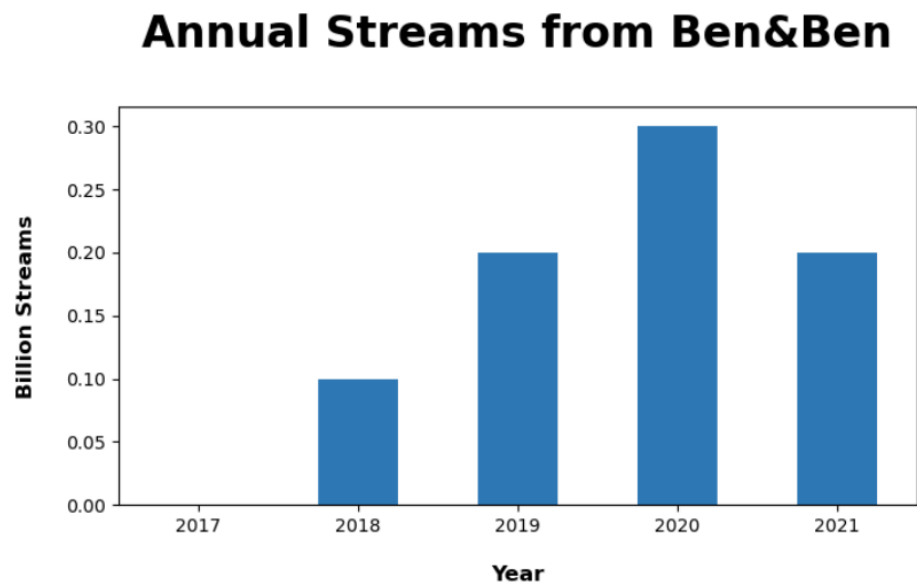


Figure 14. Annual Streams from Ben&Ben

Ben&Ben's most successful year was 2020 with a little bit over 300 million streams. They had 19 songs in the top 200 and achieved 16.3 million streams for every song.

D. Song Analysis

GLOBALLY

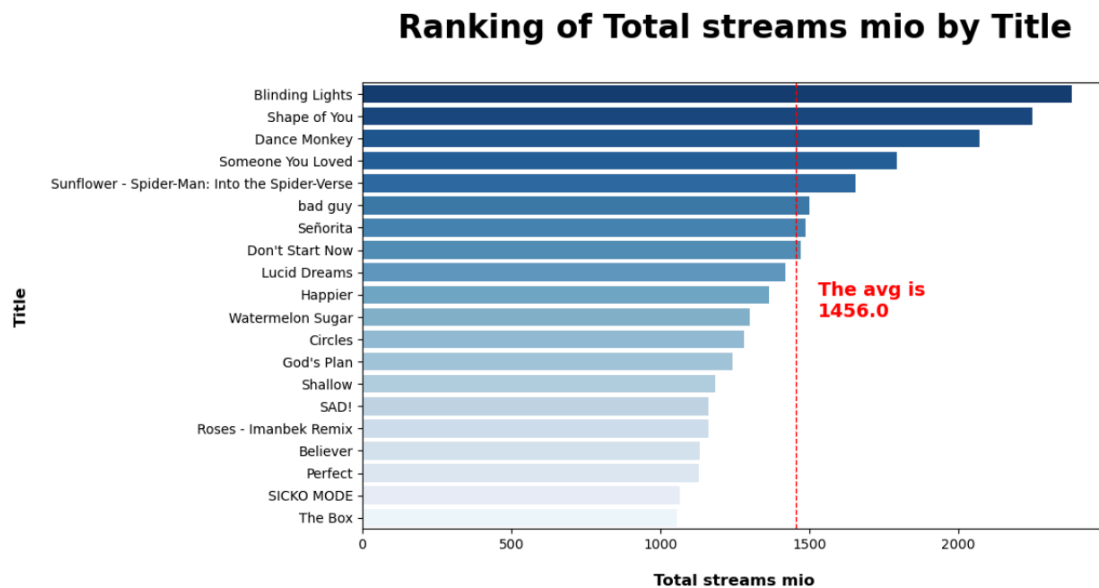


Figure 15. Ranking of Total Streams in Millions by Title of All Songs Globally

The most streamed song globally is "Blinding Lights" by the Weeknd with 2.38 billion streams. "DAKITI" does not belong to the top 20 most streamed songs. "The Weeknd" is the 10th most streamed artist with 6.8 billion streams, and DAKITI is on the 30th rank.

Avg Rank of Blinding Lights by Region

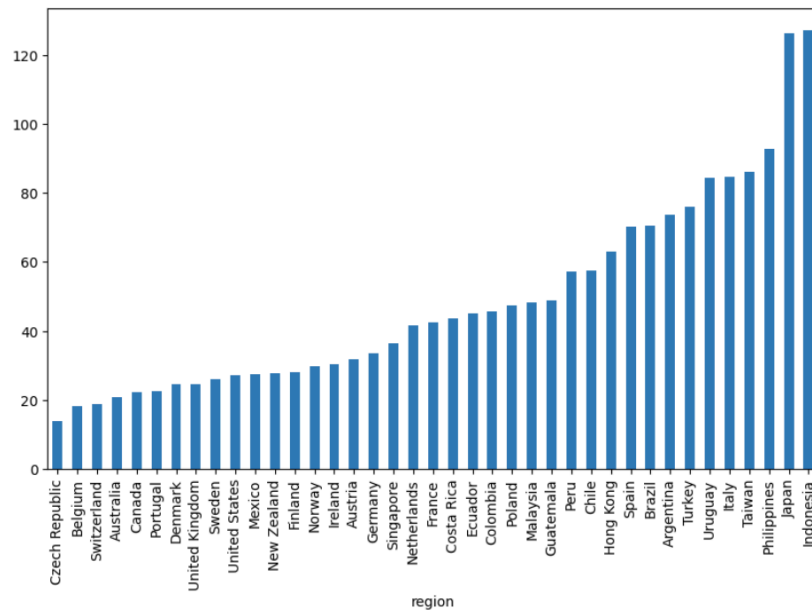


Figure 16. Average Rank of Blinding Lights by Region

Rank Development of Blinding Lights

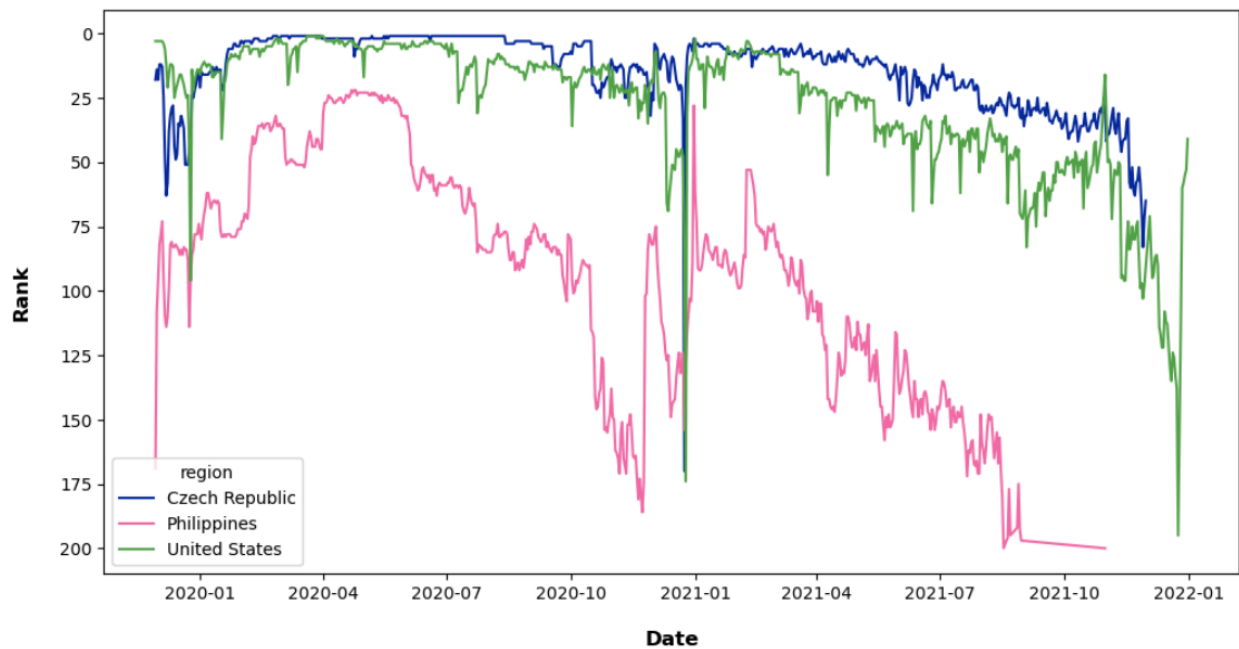


Figure 17. Rank Development of Blinding Lights in Three Relevant Regions

Blinding Lights is performing best in the Czech Republic with an avg rank of 13.9.. The duration that 'Blinding Lights' stayed in the charts in Czech Republic (in days) is 732.

LOCALLY

Ranking of Total streams mio by Title

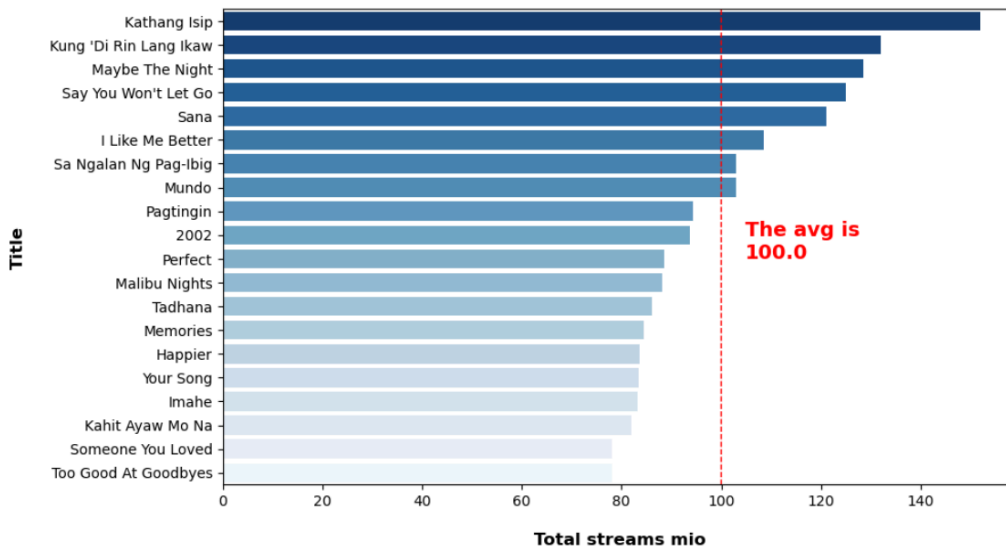


Figure 18. Ranking of Total Streams in Millions by Title of All Songs in PH

"Kathang Isip" is also on the 1st rank. "Ben&Ben", as have visualized earlier, is the most streamed artist in PH with almost 800 million streams.

Rank Development of Kathang Isip

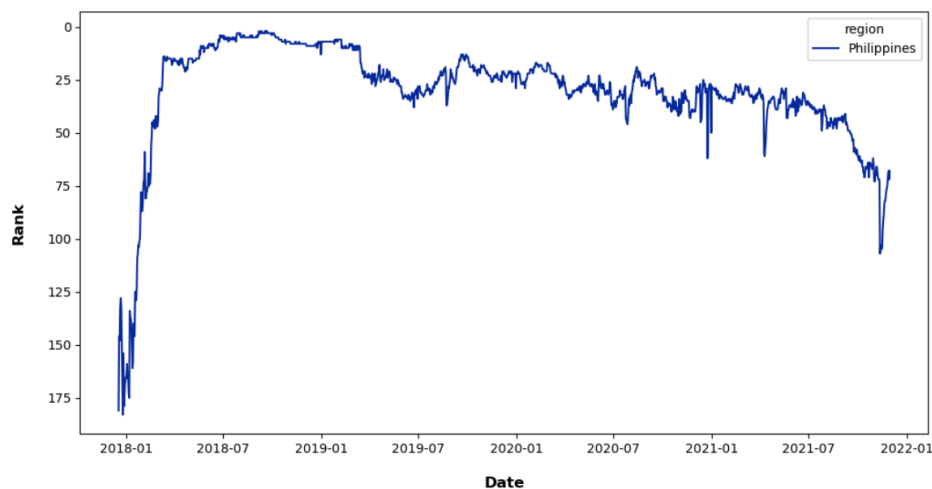


Figure 19. Rank Development of Kathang Isip

Top song of the best artist and also the top song locally. "Kathang Isip" achieves the best avg ranking in PH with ~1 and stayed for 1443 days in the charts.

Scraping Functions

```
378]: import requests

#Kyte's client ID and client secret

client_id = 'ac1a02fca6f74cea86dbc4ff719c28a3'
client_secret = '38e5e9aae2ec4722ba265438573e70a2'

380]: #To get access to spotify
def get_access_token():
    auth_url = "https://accounts.spotify.com/api/token"
    auth_response = requests.post(auth_url, {
        'grant_type': 'client_credentials',
        'client_id': client_id,
        'client_secret': client_secret,
    })
    auth_response_data = auth_response.json()
    return auth_response_data['access_token']

#To extract the id from the url
def extract_track_id(spotify_url):
    return spotify_url.split("/track/")[1].split("?")[0]

#To get track details by id
def get_track_details(track_id):
    access_token = get_access_token()
    headers = {
        'Authorization': f'Bearer {access_token}',
    }
    track_url = f'https://api.spotify.com/v1/tracks/{track_id}'
    response = requests.get(track_url, headers = headers)
    return response.json()

def get_artist_details(artist_id):
    access_token = get_access_token()
    headers = {
        'Authorization': f'Bearer {access_token}',
```

Figure 20. Screenshot of Function Code of Data Scraping

One of the recommendations for the results is to have an additional scraping for more key insights as one column of the raw dataset includes the url of the songs.

V. Summary

Spotify's global streaming volume steadily grew until 2021, which marked the first decline. The USA remains the largest market, though its share is decreasing, with Brazil and Mexico emerging as growth leaders. Globally, Bad Bunny was the most streamed artist (15.5 billion streams), with his dominance largely confined to the Americans, while "Blinding Lights" by the Weeknd was the most streamed song worldwide.

In the Philippines, Ben&Ben led the charts with nearly 800 million streams, making "Kathang Isip" the most streamed local song, which maintained the top rank for 1443 days. The data highlights the importance of localized strategies and regional market dynamics in Spotify's global growth.

With the key insights, actionable recommendations are the following:

- **Global Strategies**
 - Spotify can address the global decline in 2021 by offering exclusive content, fostering regional collaborations, and targeting growing markets such as Brazil and Mexico. Expanding the global reach of regionally dominant artists, like Bad Bunny, could enhance Spotify's penetration in untapped regions like Asia and Europe.
- **Local Strategies**
 - Localized efforts should include campaigns with Ben&Ben to reinforce their influence and spotlight Filipino talent on global platforms. Leveraging curated playlists to introduce international audiences to unique Filipino music styles could boost the global recognition of local artists.
- **Technological Enhancements**
 - Investing in real-time dashboards can help monitor artist performance and streaming trends dynamically. Additionally, scraping more detailed artist information, such as genres and audience demographics, would deepen future analyses and enable better market positioning for Spotify.